

Unit-1 LAG

- 1) Given,
- | | | |
|-----------------|---------------|---------|
| 1 st | Signal Change | = 30 s |
| 2 nd | " | = 45 s |
| 3 rd | " | = 50 s |
| 4 th | " | = 100 s |

The simultaneously Change Time = 11:50 Am.

To find:- Next Change

$$\text{LCM}(30, 45, 50, 100) \Rightarrow \begin{aligned} 30 &= 2 \times 3 \times 5 \\ 45 &= 3^2 \times 5 \\ 50 &= 2 \times 5^2 \\ 100 &= 2^2 \times 5^2 \end{aligned}$$

$$\text{LCM} = 2^2 \times 3^2 \times 5^2 = 900 \text{ Seconds} \Rightarrow \frac{900}{60} = 15 \text{ min.}$$

$\therefore$ The next Change will be $\Rightarrow 11:50 \text{ AM} + 15 \text{ min} \Rightarrow 12:05 \text{ PM}$

- 2) 1st wheel $\Rightarrow 25 \text{ rpm}$
 2nd wheel $\Rightarrow 21 \text{ rpm}$
 3rd wheel $\Rightarrow 16 \text{ rpm}$

$$\begin{aligned} \text{1st wheel revolutions in terms of seconds} &= \frac{60}{25} = \frac{12}{5} \text{ sec} \\ \text{2nd wheel} &= \frac{60}{21} = \frac{20}{7} \text{ sec} \\ \text{3rd wheel} &= \frac{60}{16} = \frac{15}{4} \text{ sec} \end{aligned}$$

$$\begin{aligned} \text{LCM} &= \frac{\text{LCM of Numerator}}{\text{HCF of Denominator}} = \frac{(2^2 \times 3)(2^2 \times 5)(3 \times 5)}{1} \\ &= \frac{2^2 \times 3 \times 5}{1} = \frac{60}{1} = 60 \text{ sec.} \end{aligned}$$

$\therefore$ They will again touch simultaneously at 60 sec / 1 min

Let the original number be " x "

$$\Rightarrow '7' \text{ is added } \Rightarrow x+7$$

$$\text{multiplied by } 5 \Rightarrow 5(x+7)$$

$$\text{divided by } 3 \Rightarrow \frac{5(x+7)}{3}$$

$$4 \text{ subtracted from quotient } \Rightarrow \frac{5(x+7)}{3} - 4$$

Resulting value is 16.

$$\frac{5(x+7)}{3} - 4 = 16$$

$$\frac{5x+35}{3} = 20 \Rightarrow 5x+35 = 60$$

$$5x = 60 - 35$$

$$x = \frac{25}{5} = 5$$

$$\boxed{x=5}$$

$\therefore$ original number is "5".

4) LCM of 8, 4, 11, 12

$$8 \Rightarrow 2^3$$

$$4 \Rightarrow 2^2$$

$$11 \Rightarrow 11$$

$$12 \Rightarrow 2^2 \times 3$$

$$\text{LCM} = 2^3 \times 3^2 \times 11$$

$$\boxed{\text{LCM} = 792}$$

Number of students must leave a remainder 3 when divided by these numbers.

$$\text{i.e. } (792 \times k) + 3$$

$$\text{when } k=1 \Rightarrow 795 \text{ (divisible by 15, but less than 1000)}$$

$$\text{when } k=2 \Rightarrow 1587 \text{ (divisible by 15, more than 1000)}$$

$\therefore$ The minimum no. of students are $\Rightarrow 1587$

Ques 1 $\Rightarrow$ 24 sec
 Ques 2 $\Rightarrow$ 36 sec
 Ques 3 $\Rightarrow$ 48 sec.

LCM $\Rightarrow$ 24, 36, 48

$$24 \Rightarrow 2 \times 2 \times 2 \times 3 = 2^3 \times 3$$

$$36 \Rightarrow 2^2 \times 3^2$$

$$48 \Rightarrow 2^4 \times 3$$

$$LCM = 2^4 \times 3^2$$

$$\Rightarrow 16 \times 9 \Rightarrow 144 \text{ seconds}$$

$$144 \text{ seconds} \Rightarrow \frac{144}{60} = 2 \text{ min } 24 \text{ sec}$$

$$\begin{array}{r} 24 \\ 60 \overline{) 144} \\ \underline{-120} \\ 240 \\ \underline{-240} \\ 0 \end{array}$$

Cost for 4 bag & 3 box = 555 Rs

Cost for 3 bag & 4 box = 460 Rs.

Cost of bag = ?

let, x be cost of bag

y be cost of box

$$4x + 3y = 555 \text{ --- (1)}$$

$$3x + 4y = 460 \text{ --- (2)}$$

from (1) & (2)

$$4 \times (1) \text{ \& } 3 \times (2) \Rightarrow 16x + 12y = 2220$$

$$\begin{array}{r} 16x + 12y = 2220 \\ - (3x + 4y = 460) \\ \hline 13x + 8y = 1760 \end{array}$$

$$7x = 840$$

$$x = \frac{840}{7} \Rightarrow \boxed{120 = x}$$

$$3(120) + 4y = 460$$

$$360 + 4y = 460$$

$$4y = \frac{460 - 360}{4} = \frac{100}{4} = \boxed{25 = y}$$

$\therefore$ Cost of bag = 120 Rs.

7) 4 Student in Each bench $\Rightarrow$ 3 bench occupied
3 Student in Each bench $\Rightarrow$ 3 students left out.
Let number of students be 'x'

$$\Rightarrow 4(x-3) = (3x+3)$$

$$4x-12 = 3x+3$$

$$\boxed{x = 15}$$

$\therefore$ The number of students = 15

8) Total people = 70

Coffee = 37

Tea = 52

To find:- Both Coffee & Tea

Total Number of people who like are $\Rightarrow 37+52$
 $= 89$.

Total people in group = 70

$$89-70 = 19$$

$\therefore$ 19 people like both Tea & Coffee

9) Let x be boys

Let y be girls

Now,

$$\frac{1}{3}x = 100 \Rightarrow x = 300$$

~~100~~ Total people participating = 300

Then, girls are $300 - 100 = 200$

$$\frac{1}{2}y = 200$$

$$\boxed{y = 400}$$

$\therefore$ 300 boys & 400 girls

Total people = $400 + 300$
 $= 700$ Students

$$\begin{aligned}
 10) & \sqrt{10 + \sqrt{25 + \sqrt{108 + \sqrt{154 + \sqrt{225}}}}} \\
 & \Rightarrow \sqrt{10 + \sqrt{25 + \sqrt{108 + \sqrt{154 + 15}}}} \\
 & \Rightarrow \sqrt{10 + \sqrt{25 + \sqrt{108 + \sqrt{164}}}} \Rightarrow \sqrt{10 + \sqrt{25 + \sqrt{108 + 13}}} \\
 & \Rightarrow \sqrt{10 + \sqrt{25 + \sqrt{121}}} \\
 & \Rightarrow \sqrt{10 + \sqrt{25 + 11}} \\
 & \Rightarrow \sqrt{10 + \sqrt{36}} = \sqrt{10 + 6} = \sqrt{16} = \textcircled{4}
 \end{aligned}$$

Unit-2 LAG

1) Given,

Marked Price of Article = 10000

Discount 1 = 10%

Discount 2 = 20%

To find: Selling price

$$1^{\text{st}} \text{ discount} = 10\% \Rightarrow \frac{10000 \times 10}{100} = 1000 \text{ Rs.}$$

$$\text{and discount} = 20\% \Rightarrow \frac{9000 \times 20}{100} = 1800$$

$$\Rightarrow 1800$$

$$\Rightarrow 9000 - 1800$$

$$= 7200$$

$\therefore$ The Selling price is 7200 Rs.

2) Total population = 12,500

1st year = Increases 10%

2nd year = decreases 15%

3rd year = Increases 20%

$$12500 \times \frac{10}{100} = 1250$$

$$12500 \times \frac{15}{100} = 1875$$

$$1^{st} \text{ year} \Rightarrow 12,500 + \left(\frac{12500 \times 10}{100} \right)$$
$$= 12,500$$

$$1^{st} \text{ year} = 10\% \text{ increase} \Rightarrow 1,250$$

$$2^{nd} \text{ year} = 15\% \text{ decrease} \Rightarrow -1,875$$

$$3^{rd} \text{ year} = 20\% \text{ increase} \Rightarrow +2,500$$

$$\therefore 12,500 + 1250 = 13,750$$

$$13,750 - 1,875 = 11,875$$

$$11,875 + 2,500 = 14,375$$

The total population after 3 yrs is 14,375.

3) Given,

Failed in Math = 35%

Failed in English = 24%

Failed in both subjects = 17%

To find:- Percentage of students passed in both

$$\Rightarrow \% \text{ of students failed in one subject is} \Rightarrow 35\% + 24\% - 17\%$$
$$\Rightarrow 42\%$$

$$\Rightarrow \% \text{ of students passed in both will be} \Rightarrow 100\% - 42\%$$
$$= 58\%$$

4) Successive % formula $\Rightarrow \left(x + y + \frac{xy}{100} + \frac{xy + yz + zx}{100} + \frac{xyz}{100 \times 100} \right)$

10%; 20%; 30%

$$\Rightarrow 10 + 20 + 30 + \frac{10(20) + 20(30) + 30(10)}{100} + \frac{10 \times 20 \times 30}{100 \times 100}$$

$$\Rightarrow 60 + \frac{200 + 600 + 300}{100} + \frac{6000}{100 \times 100}$$

$$\Rightarrow 60 + \frac{1100}{100} + \frac{6}{10} \Rightarrow 71.6\%$$

- 5) Failed in math = 35%
 " " English = 25%
 " " Hindi = 45%
 " " Math & English = 10%
 " " English & Hindi = 20%
 " " Math & Hindi = 30%
 " " all three = 5%

To find:- % of students passed in all three subjects

$$\begin{aligned}\text{Total failed} &= (35 + 25 + 45) - (10 + 20 + 30) + 5 \\ &= 105 - 60 + 5 \\ &= 50\% \text{ (failed)}\end{aligned}$$

$\therefore$ 50% will pass

6) Given,

3 tables & 12 Chairs = 2400 Rs

Tables at profit of 20%

Chairs at profit of 10%

Total profit made = 300 Rs.

Let cost of tables = x

Let cost of chairs = y

$$3x + 12y = 2400$$

$$\text{Total profit} \Rightarrow 0.20(3x) + 0.10(12y) = 300$$

$$\Rightarrow 0.6x + 1.2y = 300$$

$$6x + 12y = 3000$$

$$(6x + 12y) - (3x + 12y) = 3000 - 2400$$

$$3x = 600$$

$$\boxed{x = 200}$$

$$y = 3(200) + 12y = 2400$$

$$= 600 + 12y = 2400$$

$$12y = 1800$$

$$\boxed{y = 150}$$

$\therefore$ The cost of table, chairs is 200, 150 Rs

7) let the value of commodity be x .

$$1^{\text{st}} \text{ part} = \frac{2}{3} \times 30\% \text{ of } x \Rightarrow 20\% \text{ of } x$$

$$\text{2nd part} \Rightarrow \frac{1}{4} \times 16\% \text{ of } x = 4\% \text{ of } x$$

$$3^{\text{rd}} \text{ part} \Rightarrow \frac{1}{12} \times 12\% \text{ of } x = 1\% \text{ of } x.$$

$$\text{Total profit \%} = 20\% + 4\% + 1\% \\ = 25\%$$

Given,

$$\text{Profit} = 75 \text{ Rs.}$$

$$25\% \text{ of } 75 = \frac{25}{100} \times 75 = 75$$

$$\frac{x}{4} = 75$$

$$\boxed{x = 300}$$

$\therefore$ Value of Comodity is 300 Rs.

8) Initial investment = 5,00,000

$$1^{\text{st}} \text{ year} = \text{loss of } 4\%$$

$$\text{2nd year} = \text{profit of } 5\%$$

$$3^{\text{rd}} \text{ year} = \text{profit of } 10\%.$$

Total net profit earned is

$$1^{\text{st}} \text{ year} \Rightarrow 4\% \text{ of } 5,00,000 = 20,000 \text{ (loss)}$$

$$\text{Value in } 1^{\text{st}} \text{ year} = 4,80,000$$

$$\text{2nd year} \Rightarrow 5\% \text{ of } 4,80,000 = 24,000 \text{ (profit)}$$

$$\text{Value in 2nd year} = 4,80,000 + 24,000 \\ = 5,04,000$$

$$3^{\text{rd}} \text{ year} \Rightarrow 10\% \text{ of } 5,04,000 \Rightarrow 50,400 \text{ (profit)}$$

$$\text{Value in } 3^{\text{rd}} \text{ year} = 5,04,000 + 50,400 \\ = 5,54,400$$

$$\text{Net profit} = 5,54,400 - 5,00,000 \\ = \underline{\underline{54,400}}$$

- 9) Original cost of ghee = 100 / kg
 Vegetable oil cost = 50 / kg.
 mixture cost = 96 / kg $\Rightarrow C.P. = \frac{96}{1+0.20} = \frac{96}{1.2} = 80/kg$
 profit = 20%
 Difference b/w Ghee & Vegetable oil = 100 - 80 = 20 Rs.
 Difference b/w mixture & oil = 80 - 50 = 30.
 Ratio $\Rightarrow (30 : 20)$ (Ghee : oil)
 $\therefore$ He mixes in 3:2 ratio

10) let the price be "x"

Now,

Manufacturer to Wholesale = 10% gain.

Wholesale to Retailer = 15% gain.

Retailer to Customer = 25% gain.

Therefore,

$$x \times \left(\frac{110}{100}\right) \times \left(\frac{115}{100}\right) \times \left(\frac{125}{100}\right) = 1265$$

$$x \times 1.1 \times 1.15 \times 1.25 = 1265$$

$$x \times 1.265 \times 1.25 = 1265$$

$$x \times 1.58125 = 1265$$

$$x = \frac{1265}{1.58125}$$

$$\boxed{x = 800}$$

$\therefore$ Cost is 800 for production.

Unit - 3

LAQ's

1) Given : Total sum = Rs 25,000

Interest Rate = 15% and 18%

Total Interest = Rs 4,050

Sol:
$$\text{Single Interest (S.I)} = \frac{P \times T \times R}{100}$$

Let amount at 18% be x , 15% is $(25000 - x)$

$$4,050 = \frac{(25000 - x) \times 1 \times 15}{100} + \frac{x \times 1 \times 18}{100}$$

$$15(25,000 - x) + 18x = 405,000$$

$$375,000 - 15x + 18x = 405,000$$

$$3x = 30,000 \Rightarrow \boxed{x = 10,000}$$

$\therefore$ Amount deposited at 18% = Rs 10,000

2) Given : Amount after 5 years = Rs 1,125

Amount after 8 years = Rs 1,200

Sol:
$$\text{S.I for } n \text{ years} = \text{Amount}_2 - \text{Amount}_1$$

$$\text{Principle (P)} = \text{Amount} - \text{S.I}$$

$$\text{S.I for 3 years, } 8-5 = 3$$

$$\text{S.I for 3 years} = 1,200 - 1,125 = 75$$

$$\text{S.I for 1 year} = \frac{75}{3} = 25$$

$$\text{S.I for 5 years} = 25 \times 5 = 125$$

$$\text{Principal} = 1,125 - 125 = 1000$$

3) Given: Time = 4 years

Rate decrease = 5%

Interest decrease = Rs 100

Sol:
$$\text{Sum(P)} = \frac{\text{Difference in S.I} \times 100}{\text{Time} \times \text{Difference in Rate}}$$

$$P = \frac{100 \times 100}{4 \times 8} = 500$$

4) Given: Rates - 8% (4 years), 10% (6 yrs), 12% (5 yrs)

Total S.I after 15 years = Rs 12,160

Sol:
$$\text{Total S.I} = \frac{P \times (R_1 T_1 + R_2 T_2 + R_3 T_3)}{100}$$

$$12,160 = \frac{P \times (8 \times 4 + 10 \times 6 + 12 \times 5)}{100}$$

$$12,160 = \frac{P(32+60+100)}{100}$$

$$12,160 = \frac{P \times 152}{100}$$

$$P = \frac{1,216,000}{152} = 8,000$$

5) Given: Part 1 = Rs 12,000 at 10%.

Part 2 = x at 20%.

Average Rate = 14%

Sol: Total Interest = Total Investment $\times$ Average Rate

$$\text{Interest from Rs 12,000} = 12,000 \times \frac{10}{100} = 1200$$

$$\text{Let Total investment be } y, \text{ interest} = \frac{14}{100} y = 0.14y$$

$$\text{Part 2 investment} = (y - 12,000) \text{ interest}$$

$$= (y - 12,000) \times \frac{20}{100}$$

$$= 0.20(y - 12,000)$$

$$1,200 + 0.20y - 2,400 = 0.14y$$

$$0.06y = 1,200$$

$$y = 20,000$$

Total amount invested = 20,000 Rs

6) Given: $P = \text{Rs } 10,000$, $R = 10\%$, $T = 3 \text{ years}$

Sol:
$$\text{Amount (A)} = P \left(1 + \frac{R}{100} \right)^T$$

$$A = 10,000 \left(1 + \frac{10}{100} \right)^3 = 10000 (1.1)^3$$

$$A = 10000 \times 1.331 = 13,310$$

Amount Raviraj gets = Rs 13,310

7) Given: $P = 1000 \text{ Rs}$, $T = 1 \text{ year}$, $R = 5\%$ (half-year)

Sol:
$$A = P \left(1 + \frac{R}{2 \times 100} \right)^{2T}$$

$$A = 1000 \left(1 + \frac{5}{200} \right)^2 = 1000 (1.025)^2$$

$$= 1000 \times 1.050625 = 1,050.625$$

Amount = Rs 1,050.63

8) Given: $P = \text{Rs } 6,400$, $T = 1.5 \text{ years}$, $R = 5\%$ (half-year)

Sol:
$$A = P \left(1 + \frac{R}{2 \times 100} \right)^{2T}$$

$$A = 6,400 \left(1 + \frac{5}{200}\right)^3 = 6,400 (1.025)^3$$

$$= 6,400 \times 1.07689 = 6,892.10$$

$$\text{Amount} = \text{Rs } 6,892.10$$

9) Given: $P = \text{Rs } 12,000$, $R = 10\%$, $T = 3 \text{ years}$

Sol:
$$C.I = P \left[\left(1 + \frac{R}{100}\right)^T - 1 \right]$$

$$C.I = 12,000 [(1.1)^3 - 1]$$

$$= 12,000 [1.331 - 1] = 12,000 \times 0.331$$

$$C.I = 3,972$$

Given:
10) $P = \text{Rs } 2,500$, $R = 6\%$, $T = 2 \text{ years}$

Sol:
$$C.I - S.I = P \left(\frac{R}{100} \right)^2$$

$$C.I - S.I = 2,500 \left(\frac{6}{100} \right)^2 = 2,500 \times \frac{36}{10,000}$$

$$C.I - S.I = 9$$

Difference is 9

LAQ

Unit-4

1) Given: $A = 80\%$, $B = 85\%$, $C = 75\%$

$$\text{ratio} = 8:9:20$$

$$C - A = 18,000$$

$$\text{Income} = \frac{\text{saving}}{\text{saving}\%} \times 100$$

Sol.

~~A saves 20%~~

saves = total - spend

$$A = 100 - 80 = 20\%$$

$$B = 100 - 85 = 15\%$$

$$C = 100 - 75 = 25\%$$

$$\text{Income } A = \frac{8}{20} \times 100 = 40 \text{ units}$$

$$\text{" } B = \frac{9}{15} \times 100 = 60 \text{ "}$$

$$\text{" } C = \frac{20}{25} \times 100 = 80 \text{ "}$$

$$A:B:C = 40:60:80 = 2:3:4$$

differs b/w $C = 4$ and $A = 2$

$$4 - 2 = 2 \text{ units} = 18000$$

$$1 \text{ unit} = 9000$$

$$\begin{aligned} \text{Income of } B &= 3 \times 9000 \\ &= 27,000 \end{aligned}$$

2) Given: 4 vessels of equal sizes
spirit concentration = 60%, 70%, 75%, 80%.

Sol:
$$\text{Average Conc} = \frac{\text{Sum of Cons}}{\text{No. of vessels}}$$

$$\text{Total spirit} = 60 + 70 + 75 + 80 = 285 \text{ units}$$

$$4 \text{ vessels total volume} = 4 \times 100 = 400 \text{ units}$$

$$\text{Total water} = 400 - 285 = 115 \text{ units}$$

$$\begin{aligned} \text{Ratio (spirit : water)} &= 285 : 115 \\ &= 57 : 23 \end{aligned}$$

3) Given: Investment: Sharma = 4000 Rs,
Gupta = 8000 Rs, Sinha = 6000 Rs

Time: ^{Sharma} 6 months, Gupta = 8 months, Sinha = 8 months

Total Gain = Rs 34,000

Sol:
$$\text{Profit ratio} = (\text{Invest}_1 \times t_1) : (\text{Invest}_2 \times t_2)$$

Ratio

$$\text{Sharma} = 4000 \times 6 = 24,000$$

$$\text{Gupta} = 8000 \times 8 = 64,000$$

$$\text{Sinha} = 6000 \times 8 = 48,000$$

$$24000 : 64000 : 48000$$

$$24 : 64 : 48 = 3 : 8 : 6$$

$$\text{Total ratio} = 3 + 8 + 6 = 17$$

$$\text{Guptas share} = \frac{8}{17} \times 34000$$

$$= 8 \times 2000 = 16,000 \text{ Rs}$$

4) Given: Present age of (Sita: Son) = 5:2
after 7 years = 2:1

Sol: Let present ages be
 $5x, 2x$

$$\text{Ratio of ages} = \frac{\text{Age}_1 + t}{\text{Age}_2 + t}$$

$$\frac{5x + 7}{2x + 7} = \frac{2}{1}$$

$$5x + 7 = 4x + 14$$

$$\boxed{x = 7}$$

Present age ^{Sita} = $5 \times 7 = 35$ years

Son = $2 \times 7 = 14$ years

Age of Sita when son was born,

$$35 - 14 = 21 \text{ years.}$$

5) Given: Present age

S = Suresh

D = Daughter

$$S = 2 \times D$$

after 6 years, 23:13

$$\text{Ratio} = \frac{S+6}{D+6}$$

Sol: $\frac{2x+6}{x+6} = \frac{23}{13}$

$$\text{E3 } 26x + 78 = 23x + 138$$

$$x = 20$$

Suresh present age = $2 \times 20 = 40$ years

6) Given: Total coins : 420

Ratio of values = 1Rs : 50p : 25p = 2:3:5

Sol: $\boxed{\text{No. of Coins} = \frac{\text{Total value}}{\text{Value of one coin}}}$

Let values be $2x, 2x, 5x$

$$\text{Rs} = 1 \text{ Rs} = 100 \text{ paise}$$

$$1 \text{ Rs coin} = \frac{2x}{1} = 2x$$

$$50p \quad " = \frac{2x}{0.5} = 6x$$

$$25p \quad " = \frac{5x}{0.25} = 20x$$

$$\text{Total coins} = 2x + 6x + 20x = 28x = 420$$

$$x = 15$$

No. of 1Rs coins

$$= 2 \times 15$$

$$= 30$$

7) Given: 3 jugs equal with milk : water ratios of 1:2, 2:3, 3:4

Given: Take LCM of total parts

$$(1+2, 2+3, 3+4) \Rightarrow (3, 5, 7) = 105$$

$$\text{Jug 1} = \frac{105}{3} = 35, \text{ milk} = 35 \times 1, \text{ water} = 35 \times 2 = 70$$

$$\text{Jug 2} = \frac{105}{5} = 21, \text{ milk} = 2 \times 21 = 42, \text{ water} = 3 \times 21 = 63$$

$$\text{Jug 3} = \frac{105}{7} = 15, \text{ milk} = 3 \times 15 = 45, \text{ water} = 4 \times 15 = 60$$

$$\begin{aligned} \text{Total milk} &= 35 + 42 + 45 \\ &= 122 \end{aligned}$$

$$\text{Total water} = 70 + 63 + 60 = 193$$

$$M:W = 122 : 193$$

8) Given: Total sum = 1290

$$A = 1\frac{1}{2}B, \quad B = 1\frac{3}{4}C$$

Sol:
$$C's \text{ share} = \frac{C's \text{ ratio}}{\text{Total ratio}} \times \text{Total sum}$$

$$B = \frac{7}{4}C, \quad A = \frac{3}{2}B = \frac{3}{2} \times \frac{7}{4}C = \frac{21}{8}C$$

$$A:B:C = \frac{21}{8} : \frac{7}{4} : 1 \Rightarrow 21 : 14 : 8$$

$$\text{Total parts} = 21 + 14 + 8 = 43$$

$$C's \text{ share} = \frac{8}{43} \times 1290 = 8 \times 30 = 240 \text{ Rs}$$

9) Given : Shanker = Rs 1,20,000 (12 months)
Aniket = Rs 1,90,000 (9 months)

$$\text{Total profit} = 17,50,000$$

$$\text{Profit Ratio} = (Invest_1 \times t_1) \times (Invest_2 \times t_2)$$

$$(12 \times 12) : (19 \times 9) = 144 : 171 = 16 : 19$$

$$\text{Total parts} = 16 + 19 = 35$$

$$\begin{aligned} \text{Aniket share} &= \frac{19}{35} \times 17,50,000 \\ &= 19 \times 50000 = 950000 \end{aligned}$$

10) given : $2^{\text{nd}} = \frac{2 \times 1^{\text{st}}}{3}$ ①, $2^{\text{nd}} = \frac{3 \times 3^{\text{rd}}}{2}$ ②
Average = 44

Sol: Equal the values $1 \times \textcircled{2}$, $2 \times \textcircled{1}$

$$\begin{aligned} \frac{2^{\text{nd}}}{1^{\text{st}}} &= \frac{2}{1} \times \frac{3}{3}, \quad \frac{2^{\text{nd}}}{3^{\text{rd}}} = \frac{3}{1} \times \frac{2}{2} \\ &= \frac{6}{3} \qquad \qquad \qquad = \frac{6}{2} \end{aligned}$$

Let 2^{nd} be $6x$, 1^{st} be $3x$, 3^{rd} be $2x$

$$\text{Average} = \frac{\text{Sum of numbers}}{\text{no. of values}}$$

$$\text{Average} = \frac{6x + 3x + 3x + 6x + 2x}{5} = 44$$

$$11x = 132$$

$$x = 12$$

$$\text{Number, 1st} = 3 \times 12 = 36$$

$$2\text{nd} = 3 \times 12 = 36$$

$$3\text{rd} = 2 \times 12 = 24$$

$$\text{Largest number} = 75 \rightarrow 2\text{nd}$$

Q. 2A8

- 1) Given: $S_1 = 60 \text{ km/h}$, Late $t_1 = 20 \text{ min}$
 $S_2 = 80 \text{ km/h}$, Early $t_2 = 10 \text{ min}$

Sol:
$$\text{Distance} = \frac{S_1 \times S_2}{S_2 - S_1} \times (t_1 + t_2)$$

$$t_1 + t_2 = 20 + 10 = 30 \text{ min}$$

$$30 \text{ min} = \frac{30}{60} = 0.5 \text{ hours}$$

$$\text{Distance} = \frac{60 \times 80}{80 - 60} \times 0.5 = \frac{60 \times 80}{20} \times 0.5$$

$$\text{Distance} = 120 \text{ km}$$

- 2) Given: Total Distance of P & Q = 22 km

$$S_A = 5 \text{ km/h}, S_B = 6 \text{ km/h}$$

Sol:

$$\text{Distance} = \text{Speed} \times \text{time}$$

$$\text{Distance covered by both} = 22 \times 2 = 44 \text{ km}$$

$$\text{Time} = \frac{44}{5+6} = \frac{44}{11} = 4 \text{ hours}$$

$$\text{Dist from P to R} = S_A \times T = 5 \times 4$$

$$\text{Distance} = 20 \text{ km}$$

3) Given: Speed without stoppages (S_{fast}) = 40 km/h
Speed with stoppages (S_{slow}) = 30 km/h

Sol: ~~Time of stoppage = $\frac{S_{fast} - S_{slow}}{S_{fast}}$~~

$$\text{Time of stoppage} = \frac{S_{fast} - S_{slow}}{S_{fast}}$$

$$\begin{aligned} 1) \text{ Stopping time} &= \frac{40 - 30}{40} \times 60 \\ &= \frac{10}{40} \times 60 = \frac{1}{4} \times 60 \\ &= 15 \text{ min} \end{aligned}$$

4) Given: Distance start = 100 m
 $S_p = \frac{1 \text{ km} \times 8 \text{ min}}{8} = 8 \text{ km/min} = \frac{60}{8} \text{ km/h} = 7.5 \text{ km/h}$
 $S_t = \frac{1 \text{ km} \times 10 \text{ min}}{10} = 10 \text{ km/min} = \frac{60}{10} \text{ km/h} = 6 \text{ km/h}$

Sol: $\text{Time} = \frac{\text{Initial Distance}}{\text{Relative Speed}}$

$$\text{Relative speed} = 7.5 - 6 = 1.5 \text{ km/h}$$

$$\text{initial distance} = 100 \text{ m} = 0.1 \text{ km}$$

$$\text{Time} = \frac{0.1}{1.5} = \frac{1}{15} \text{ hours}$$

$$\text{Distance} = \text{Speed} \times \text{time}$$

$$\begin{aligned} D_t &= 6 \times \frac{1}{15} \\ &= 0.4 \text{ km} \end{aligned}$$

Distance they
covers = 400 meters

5) Given: $S_1 = 3 \text{ km/h}$, $\text{Late} = 15 \text{ mins}$
 $S_2 = 4 \text{ km/h}$, $\text{Early} = 15 \text{ min}$

Sol: $\boxed{\text{Distance} = \frac{S_1 \times S_2}{S_2 - S_1} \times (t_1 + t_2)}$

Total time $= t_1 + t_2 = 15 + 15 = 30 \text{ min} = 0.5 \text{ hours}$

$$\text{Distance} = \frac{3 \times 4}{4 - 3} \times 0.5$$

$$= 12 \times 0.5 = 6 \text{ km}$$

6) Given: $S_1 = 44 \text{ km/h}$, $S_2 = 77 \text{ km/h}$

$t_1 + t_2 = 6 \text{ hours}$

Sol: $\text{Distance} = \frac{44 \times 77}{77 - 44} \times 6 = \frac{4 \times 77}{3} \times 6$

$$= 4 \times 77 \times 2 = 616 \text{ km}$$

7) Given: Total Time = 12 hr

first half speed (S_1) = 35 km/h

second half speed (S_2) = 30 km/h

Sol: $\boxed{\text{Average speed} = \frac{2 \times S_1 \times S_2}{S_1 + S_2}}$

$$\text{Average speed} = \frac{2 \times 15 \times 30}{15 + 30} = \frac{900}{45} = 20 \text{ km/h}$$

$$\begin{aligned} \text{Total Dist} &= \text{Avg speed} \times \text{total time} \\ &= 20 \times 12 = 240 \text{ km} \end{aligned}$$

8) Given: Walk + Ride = 6 hr 30 min
 Ride + Ride = (6 hr 30 min) - (2 hr 10 min)
 $= 4 \text{ hr } 20 \text{ min}$

Sol: $2 \times (\text{Walk} + \text{Ride}) = 13 \text{ hours}$

$$\text{Ride} + \text{Ride} = 4 \text{ hr } 20 \text{ min}$$

Time for walking both ways

$$= 13 \text{ hr} - 4 \text{ hr } 20 \text{ min}$$

$$= 8 \text{ hr } 40 \text{ min}$$

9) Given: $V_t = 35 \text{ km/h}$, $V_m = 5 \text{ km/h}$ (same direction)

Time to pass platform $t_p = 18 \text{ sec}$, Time to pass man $t_m = 12 \text{ sec}$

Sol: length of train = Relative speed $\times$ time to pass

$$\text{Relative speed} = 35 - 5 = 30 \text{ km/h} = 30 \times \frac{5}{18} = \frac{25}{3} \text{ m/s}$$

$$\text{Train length (L)} = \frac{25}{3} \times 12 = 100 \text{ m}$$

$$\text{Train speed} = 35 \times \frac{5}{18} = \frac{175}{18} \text{ m/s}$$

$$\text{Total Dist} = \text{speed} \times \text{time}$$

$$= \frac{175}{18} \times 18 = 175 \text{ m}$$

$$\text{Platform length} = 175 - 100$$
$$= 75 \text{ m}$$

$$\text{Train} = 100 \text{ m}, \text{ Platform} = 75 \text{ m}$$

10) Given: $t_u = 10 \text{ hr}$, $t_D = 7 \text{ hr}$

$$\text{Ratio (Boat: Current)} = \frac{t_u + t_D}{t_u - t_D}$$

Sol: Ratio = $\frac{10+7}{10-7}$

$$= \frac{17}{3}$$

$$\text{Ratio} = 17:3$$